

Introduction to Advanced Exploits

Zero-Day Exploits

A zero-day exploit is a vulnerability in software that is unknown to the vendor and doesn't yet have a patch. Hackers can exploit it before developers even realize the flaw exists. These are extremely dangerous because:

- There's no fix available initially.
- They're often sold on black markets.
- Can be used in targeted attacks (e.g., cyber-espionage or APTs).

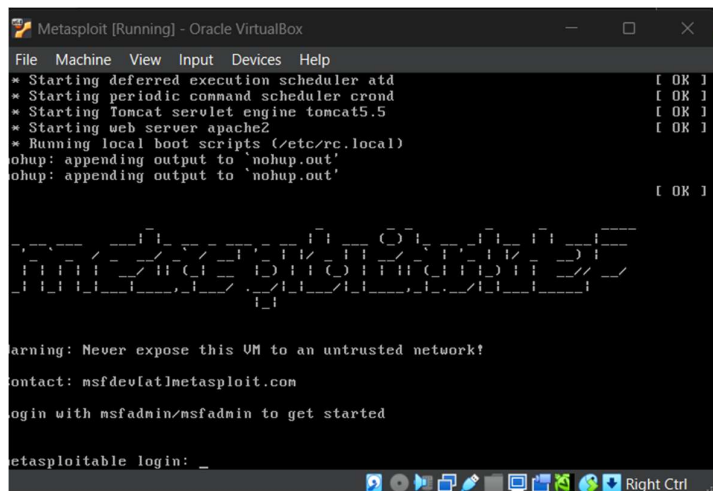
Understanding Buffer Overflow Attacks

A buffer overflow happens when a program writes more data to a buffer (temporary storage) than it can handle. This excess data can overwrite adjacent memory and Attackers exploit buffer overflows to inject shellcode, hijack control of the program counter (EIP), and redirect execution flow to their malicious payload.

- It crash the program,
- It allow execution of malicious code.

Example: Overwriting the return address on the stack with the address of injected shellcode.

For metasploitable2 its already done in previous weeks



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Metasploit [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
* Starting deferred execution scheduler atd [ OK ]
* Starting periodic command scheduler crond [ OK ]
* Starting Tomcat servlet engine tomcat5.5 [ OK ]
* Starting web server apache2 [ OK ]
* Running local boot scripts (/etc/rc.local)
ohup: appending output to 'nohup.out'
ohup: appending output to 'nohup.out' [ OK ]

Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started

metasploitable login: _
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